

Assignment 13 - AI for Undefined Behaviour Detection in C/C++ Programs

Project Problem Description

Study whether an LLM can detect undefined behaviour (UB) in C/C++ source code before LLVM/Clang optimizations exploit that UB and transform the program in surprising ways.

In C/C++, operations such as signed integer overflow, null dereference, invalid shifts, and out-of-bounds access may have undefined behaviour. LLVM can legally optimize under the assumption that UB does not occur, which can change program behaviour in surprising ways.

Objective: Evaluate whether an LLM can (a) identify UB patterns, (b) explain UB causes, (c) predict LLVM optimization effects, and (d) compare with UBSan, clang-tidy, and static analyzers.

Overall Completion Percentage

Project completion: **100%** for assignment deliverables.

What Has Been Done (Non-Technical Summary)

- Prepared a full benchmark set of UB examples, plus safe examples for false-positive checking.
- Ran compiler experiments to show how optimization changes behaviour or reasoning.
- Collected outputs from UBSan, clang warnings, clang-tidy, and cppcheck.
- Collected LLM analysis results for all benchmark files.
- Built final comparison table and summary percentages.
- Wrote final report and generated PDF.
- Prepared final presentation and generated PPTX.
- Added a user-facing UI for uploading a file and running 5-way comparison (UBSan, Clang warnings, clang-tidy, cppcheck, LLM).

Deliverables Status

- Benchmark set of UB programs: Completed.
- LLVM/Clang optimization experiment artifacts: Completed.
- LLM vs tool comparison: Completed.
- Report with soundness/completeness/FP/FN discussion: Completed.
- Final presentation with hybrid pipeline proposal: Completed.

Is Anything Left?

No required assignment deliverable is pending. Optional polish items (not required) include visual refinement of slides and optional cleanup of accidental/unneeded files.

Final Readiness

The project is ready for submission according to the provided Assignment 13 description and deliverables.